

```

1 import java.util.Scanner;
2
3 public class UserInfoLab {
4     public static void main(String[] args) {
5         // Part 1
6         // Create a Scanner for keyboard input
7         // Ask the user to enter their first and last name and
        pass these
8         // values to the generateUsername method and save the
        returned result.
9         Scanner scan = new Scanner(System.in);
10        String first, last;
11
12        System.out.print("Can you please enter your first name?\t"
13    );
14        first = scan.nextLine();
15
16        System.out.print("Can you please enter your last name?\t"
17    );
18        last = scan.nextLine();
19
20        System.out.println("Username: "+ generateUsername(first,
21    last)+ '\n');
22        String user = generateUsername(first, last);
23        // Part 2
24        // Ask the user to enter a password and pass this value to
        the validatePassword method.
25        // The validatePassword method will check if the password
        meets the criteria:
26        String password;
27
28        System.out.print("Can you please enter your password?\t");
29
30        password = scan.nextLine();
31        boolean valPass = validatePassword(password);
32        // Part 3
33        // If the user entered a valid password in step 2, then
        ask the user to enter their
34        // credit card number and pass this value to the
        maskCreditCard method.
35        String creditNum = "";
36        if(valPass){
37            System.out.print("What is your credit card number?\t"
38    );
39            creditNum = scan.nextLine();
40        }

```

```

37         creditNum = maskCreditCard(creditNum);
38         // Part 4
39         // If the user entered a valid password AND valid credit
card number, display the output
40         // as shown in the demo video
41         // https://drive.google.com/file/d/
1sM0w5wk0gSfuUcvQhFyZ5flnv_d9qQd3/view?usp=sharing
42         if(!(creditNum.equals("N/A"))&& valPass){
43             System.out.println("\nFinal details:");
44             System.out.println("Username: \t" + user);
45             System.out.println("Credit card: \t" + creditNum);;
46         }
47
48     }
49
50     public static String generateUsername(String firstName, String
lastName) {
51         String result = "";
52         if(firstName.length() < 3|| lastName.length() < 3){
53             if(firstName.length() < 3){
54                 result += firstName;
55             }
56             else{
57                 result += firstName.substring(0, 3);
58             }
59             if(lastName.length()<3){
60                 result += lastName;
61             }
62             else{
63                 result += lastName.substring(0, 3);
64             }
65         }
66         else{
67             result += firstName.substring(0, 3);
68             result += lastName.substring(0, 3);
69         }
70
71         result = result.toLowerCase();
72         return result;
73     }
74     public static boolean validatePassword(String password) {
75         int count=0;
76         boolean contUpper = true, eightLong = true;
77         boolean contDig = containsDigit(password);
78
79         for(int i = 0; i< password.length(); i++){

```

```

80         String letter= password.substring(i,i+1);
81         if(letter.equals(letter.toUpperCase())){
82             count++;
83         }
84     }
85
86     if(password.length() < 8) {
87         eightLong = false;
88         System.out.println("Password is not 8 characters long
89         .");
90     }
91     if(count == 0){
92         contUpper = false;
93         System.out.println("Password does not contain an
94         Upper Case Letter.");
95     }
96     if(!contDig){
97         System.out.println("Password does not contain a digit
98         .");
99     }
100
101     if(!(contDig && contUpper && eightLong)){
102         return false;
103     }
104     else{
105         System.out.println("Valid password. Checking Credit
106         Card");
107     }
108     return true;
109 }
110
111 public static String maskCreditCard(String creditCardNumber
112 ) {
113     String result = "";
114     if(allDigits(creditCardNumber) && creditCardNumber.length
115     () == 16){
116         for(int i = 1; i<= 3; i++){
117             result += "**** ";
118         }
119         result += creditCardNumber.substring(12);
120     }
121     else{
122         result = "N/A";
123     }
124     return result;
125 }
126
127 }
128
129

```

```
120    /**
121        This method verifies that the string contains at least one
numeric digit
122        @param str The string to check
123        @return true or false if a digit is present
124    */
125    public static boolean containsDigit(String str) {
126        char[] chars = str.toCharArray();
127        for (char c: chars) {
128            if (Character.isDigit(c))
129                return true;
130        }
131        return false;
132    }
133
134    /**
135        * Checks if the entire String is all numerical
136        * @param str The string to check
137        * @return true or false if the string is ALL digits
138    */
139    public static boolean allDigits(String str) {
140        char[] chars = str.toCharArray();
141        for (char c: chars) {
142            if (!Character.isDigit(c))
143                return false;
144        }
145        return true;
146    }
147
148 }
149
```